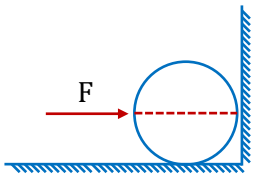
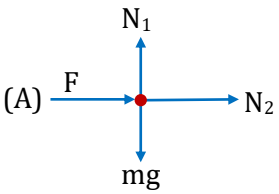
EXERCISE (O-1)

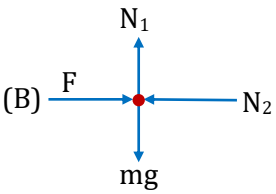
SINGLE CORRECT TYPE QUESTIONS

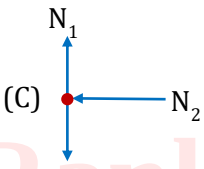
Newton's Laws of Motion

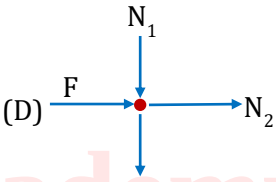
1. A ball of mass m kept at the corner as shown in the figure, is acted by a horizontal force F. The correct free body diagram of ball is



(A) 

(B) 

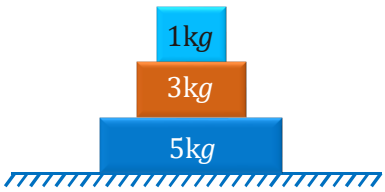
(C) 

(D) 

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PNL173

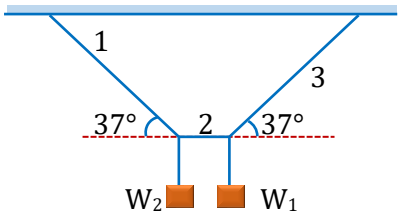
2. All three blocks placed on horizontal surface at rest than net force on the middle block is?



(A) 30 N (B) 40 N (C) 90 N (D) 0N

PNL078

3. In a given figure system is in equilibrium. If W_1 = 300 N. Then W_2 is approximately equal to

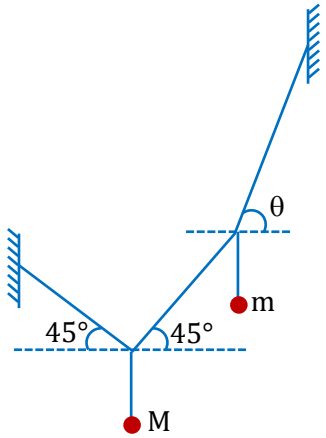


(A) 500 N (B) 400 N (C) 670 N (D) 300 N

PNL076

For More Join: @IITJEE_Advanced

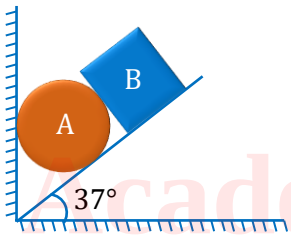
4. Two masses m and M are attached to strings as shown in the figure. In equilibrium tan θ is :



(A) $1 + (2M/m)$ (B) $1 + (2m/M)$ (C) $1 + (M/2m)$ (D) None of these

PNL176

5. A perfect smooth sphere A of mass 2 kg is in contact with a rectangular block B of mass 4 kg and vertical wall as shown in the figure. All surfaces are smooth. Find normal reaction by vertical wall on sphere A

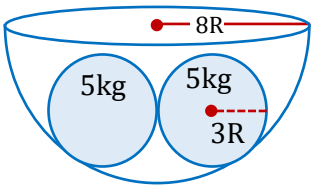


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(A) 20 N (B) 25 N (C) 80 N (D) 45 N

PNL080

6. In a hemispherical bowl of radius $8R$, two balls of mass 5 kg are placed in the equilibrium. Radius of the ball is $3R$. Normal contact force between the two balls is given by



(A) $\frac{75}{2}$ (B) $\frac{75}{4}$ (C) $\frac{75}{6}$ (D) 15

PNL177

7. Three identical blocks each of mass $20kg$ are connected by ideal strings and are being pulled along a horizontal frictionless table by a constant horizontal force F as shown in figure. What is the magnitude of force F if the tension in the string connecting blocks A and B is 60 N ?

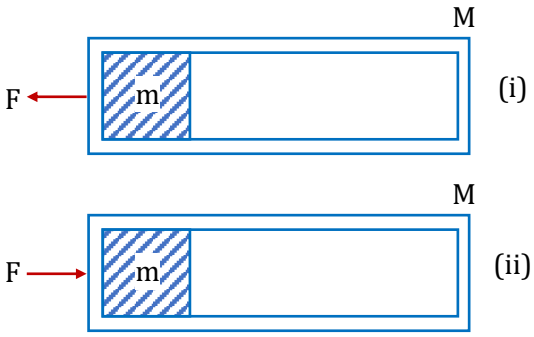


(A) 60 N (B) 120 N (C) 180 N (D) 90 N

PNL178

For More Join: @IITJEE_Advanced

8. A block of mass m is placed in contact with one end of a smooth tube of mass M (see figure). A horizontal force F acts on the tube in each case (i) and (ii). Then mark **incorrect** option :



(A) $a_m = 0$ and $a_M = \frac{F}{M}$ in (i)

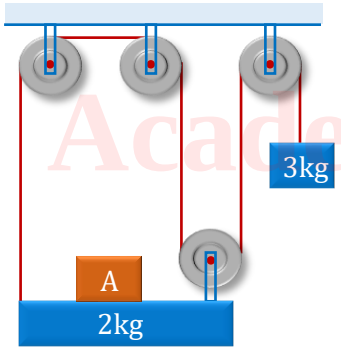
(B) $a_m = a_M = \frac{F}{M+m}$ in (i)

(C) $a_m = a_M = \frac{F}{M+m}$ in (ii)

(D) Force on m is $\frac{mF}{M+m}$ in (ii)

PNL179

9. The arrangement shown in the figure is in equilibrium. What is the contact force (in N) between the block A and 2 kg block ? ($g = 10\ m/s^2$)

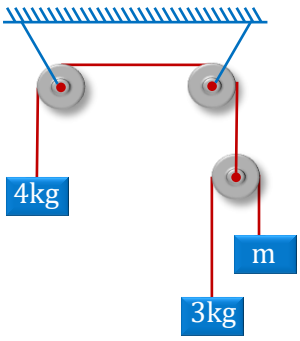


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(A) 70 N (B) 100 N (C) 30 N (D) 50 N

PNL126

10. What should be the value of "m" so that 4kg block remains at rest :-

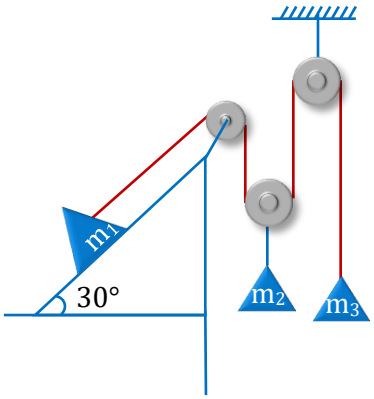


(A) 3 kg (B) 4 kg (C) 1.5 kg (D) 2.5 kg

PNL180

For More Join: @IITJEE_Advanced

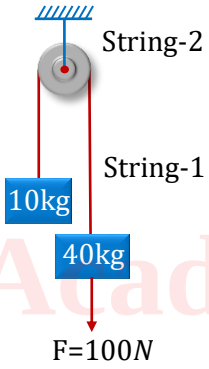
11. All pulleys and strings are massless, for what ratio of $m_1 : m_2 : m_3$ system will be in equilibrium (inclined is frictionless)



(A) 1 : 2 : 1 (B) 2 : 2 : 1 (C) 1 : 2 : 2 (D) 2 : 1 : 2

PNL181

12. Mark the **INCORRECT** statement(s) :



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Magnitude of acceleration of both blocks is 8 m/s^2

(A) tension in the string is 180 N

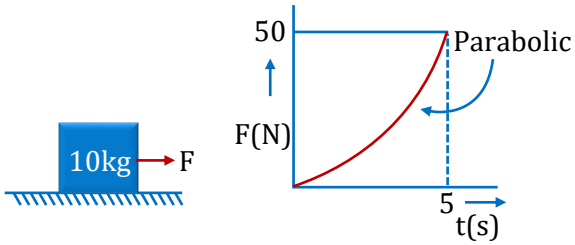
(B) acceleration of mass 10 kg is $8m/s^2$ upward

(C) acceleration of mass 10 kg is $8m/s^2$ downward

(D) acceleration of mass 40 kg is $8m/s^2$ downward

PNL115

13. A force F is applied to the initially stationary cart. The variation of force with time is shown in the figure. The speed of cart at t = 5 second is :–

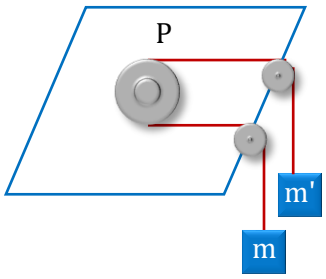


(A) 10 m/s (B) 8.33 m/s (C) 2 m/s (D) Zero

PNL182

For More Join: @IITJEE_Advanced

14. A smooth pulley P of mass $M = 2\ kg$ is lying on a smooth table. A light string passes round the pulley and has mass $m = 1kg$ and $m' = 2kg$ attached to its ends, the two portions of the string being perpendicular to the edge of the table so that the masses hang vertically. Acceleration of pulley is–

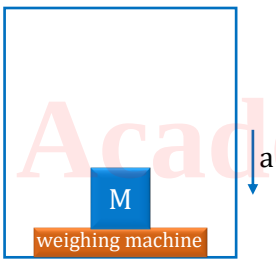


(A) $\frac{g}{2}$ (B) $\frac{2g}{3}$ (C) $\frac{4g}{9}$ (D) $\frac{4g}{7}$

PNL185

15. With what acceleration 'a' shown the elevator descends so that the block of mass M exerts a force of $\frac{Mg}{10}$ on the weighing machine? [g = Acceleration due to gravity]

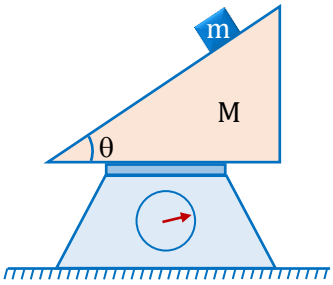
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(A) 0.3 g (B) 0.1 g (C) 0.9 g (D) 0.6 g

PNL186

16. The wedge of mass M sits at rest on a scale as shown in figure. A small block of mass m slides down the frictionless incline of the wedge. The scale reading while the block slides, is



(A) $(m + M)g$ (B) $Mg + mg\cos^2\theta$

(C) $(m + M)g + mg\sin^2\theta$ (D) $(M - m)g + mg\sin^2\theta$

PNL084

For More Join: @IITJEE_Advanced

17. A lift is going up. The total mass of the lift and the passengers is 500 kg. The speed of lift varies with time as shown in figure. What will be the tension (in kN) in the rope pulling the lift at t = 27s.

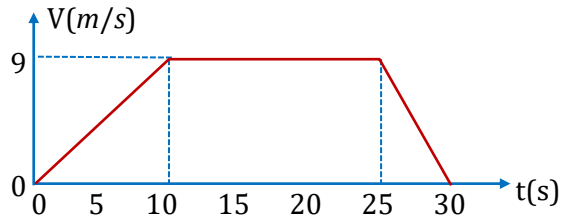
(g = 9.8 m/s^2)

(A) 2

(B) 4

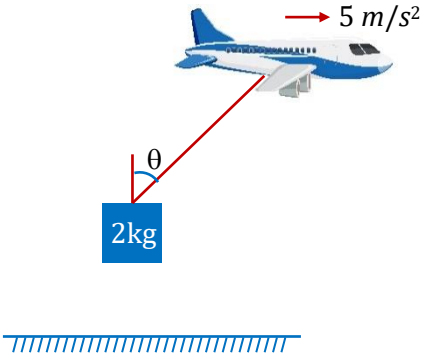
(C) 6

(D) 8



PNL110

18. An airplane is moving horizontally with constant acceleration 5 m/s^2 with a block of mass 2kg is hanging with it, which is at rest w.r.t. plane find angle θ which string makes with vertical?

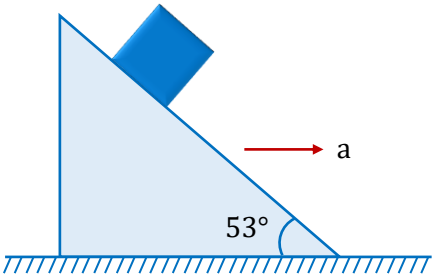


(A) tan θ = 2 (B) cot θ = 2 (C) sinθ = $\frac{1}{2}$ (D) cos θ = $\frac{1}{2}$

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PNL111

19. If acceleration (in m/s^2) of wedge (rightwards) so that block does not move w.r.t. wedge is $\frac{8n}{3}$. Value of n.



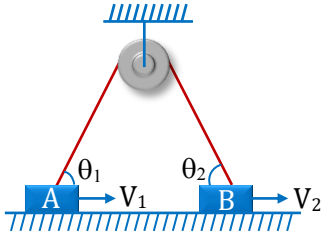
(A) 7 (B) 8 (C) 5 (D) 9

PNL116

20. In the figure shown, blocks A and B move with velocities v_1 and v_2 along horizontal direction. Find the ratio of $\frac{v_1}{v_2}$.

(A) $\frac{\cos\theta_1}{\cos\theta_2}$ (B) $\frac{\cos\theta_2}{\cos\theta_1}$

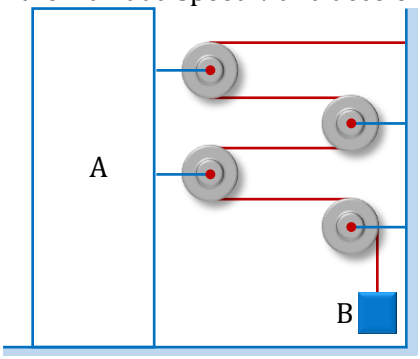
(C) $\frac{\sin\theta_2}{\sin\theta_1}$ (D) $\frac{\sin\theta_1}{\sin\theta_2}$



PNL128

For More Join: @IITJEE_Advanced

21. Block A is moving away from the wall at a speed v and acceleration a.

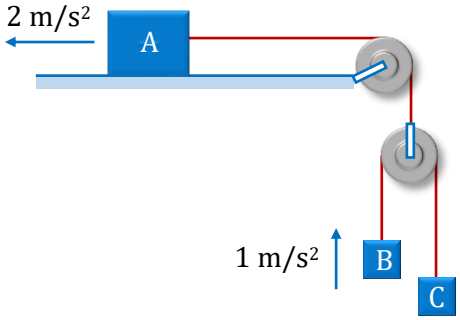


(A) Velocity of B is v with respect to A. (B) Acceleration of B is a with respect to A.

(C) Acceleration of B is $4a$ with respect to A. (D) Acceleration of B is $\sqrt{17a}$ with respect to A.

PNL265

22. In the setup shown, find acceleration of the block C.

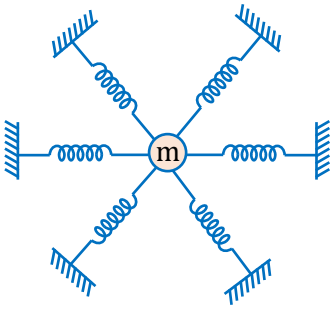


(A) 3 $m/s^2 \uparrow$ (B) 3 $m/s^2 \downarrow$ (C) 5 $m/s^2 \uparrow$ (D) 5 $m/s^2 \downarrow$

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PNL266

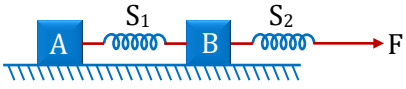
23. A sphere of mass 'm' is kept in equilibrium with the help of several springs as shown in the figure. Measurements show that one of the springs applies a force $\vec{F}$ on the sphere. With what acceleration the sphere will move immediately after this particular spring is cut?



(A) Zero (B) $\frac{\vec{F}}{m}$ (C) $-\frac{\vec{F}}{m}$ (D) Insufficient information

PNL091

24. Two blocks A and B of equal mass 2 kg are arranged as shown with springs S_1 and S_2. Spring S_2 is pulled with force $F = 6\ N$. Acceleration of block B is 4 m/s^2 towards right at an instant. If F is suddenly removed at this instant, the instantaneous acceleration of A and B will be (Assume no friction)



(A) $a_A = 1\ m/s^2$ towards right and $a_B = 4\ m/s^2$ left

(B) $a_A = 1\ m/s^2$ towards left and $a_B = 4\ m/s^2$ right

(C) $a_A = 1\ m/s^2$ towards left and $a_B = 1\ m/s^2$ right

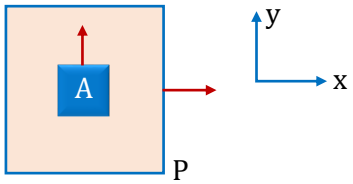
(D) $a_A = 0$ and $a_B = 4\ m/s^2$ right

PNL098

For More Join: @IITJEE_Advanced

Friction

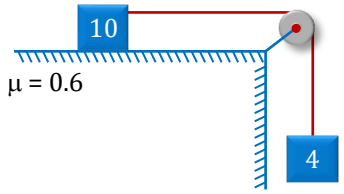
25. A block is moving along *y*-axis with velocity 4 *m/s* on a plank relative to ground and the plank is moving along *x*-axis with velocity 3 *m/s* at any instant of time. The unit vector in direction of friction force acting on block at this moment of time is.



(A) $\frac{3\hat{i}+4\hat{j}}{5}$ (B) $\frac{3\hat{i}-4\hat{j}}{5}$ (C) $\frac{-3\hat{i}+4\hat{j}}{5}$ (D) $\frac{-3\hat{i}-4\hat{j}}{5}$

PNL142

26. For the system of blocks and pulley shown in the figure, acceleration of block is "*a*" and friction force acting on 10 *kg* block is *F* then :-



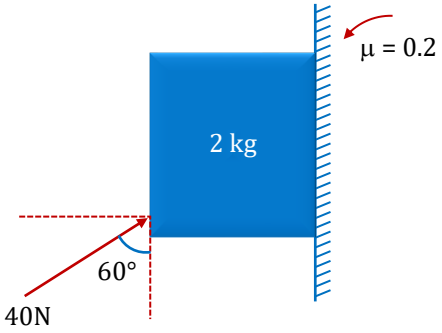
(A) $F = 60\ N, a = 0\ m/s^2$ (B) $F = 50\ N, a = \frac{20}{7}\ m/s^2$

(C) $F = 40\ N, a = 0\ m/s^2$ (D) $F = 40\ N, a = \frac{20}{7}\ m/s^2$

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PNL141

27. A block of mass 2 *kg* is pressed against the vertical wall of coefficient of friction μ = 0.2. The value of friction force acting on the wall is :-

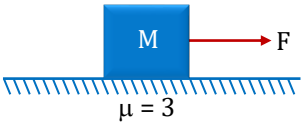


(A) $40\sqrt{3}\ N$ in upward direction (B) $40\sqrt{3}\ N$ in downward direction

(C) 20 *N* in upward direction (D) No friction force acts on the block

PNL143

28. A block of mass *M* placed on rough surface of coefficient of friction equal to 3. If *F* is the (4/5) of the minimum force required to just move, find out the force exerted by ground on the block.



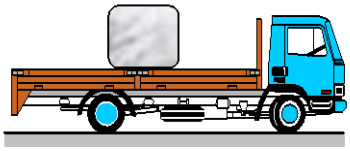
(A) 2.6 *Mg* (B) *Mg*

(C) 4 *Mg* (D) 3.4 *Mg*

PNL144

For More Join: @IITJEE_Advanced

29. A truck carries a 2 Ton block on a level horizontal road. The coefficient of static and kinetic frictions between the stone-block and truck-bed are 0.4 and 0.2 respectively. When the truck accelerates at 2 m/s^2, how much friction acts between the stone-block and the truck-bed? [g = 10 m/s^2]



(A) 2000 N (B) 4000 N (C) 8000 N (D) 10000 N

PNL189

30. The coefficient of friction between the block and the horizontal surface is μ as shown in figure. The block moves towards right under action of horizontal force F (figure-a). Sometime later another force P is applied to the block making an angle θ (such that tan θ = μ) with vertical as shown in (figure - b). After application of force P, the acceleration of block shall

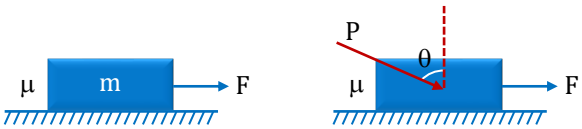


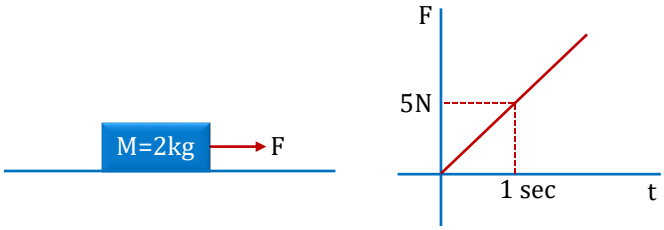
Fig. (a) **Fig. (b)**

(A) increase
(B) decrease
(C) remains same
(D) information insufficient for drawing inference

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PNL146

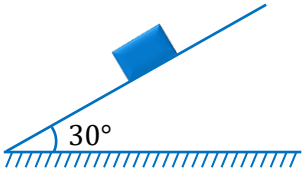
31. A block of mass 2kg is at rest on a rough surface, of coefficient of friction 0.5. A force F is applied on the block, variation of force is represented in the figure. Speed of the block at t = 4 sec is:



(A) 20 m/s (B) 10 m/s (C) 5 m/s (D) 15 m/s

PNL147

32. Figure shows a block kept on a rough inclined plane. The maximum external force down the incline for which the block remains at rest is 2N while the maximum external force up the incline for which the block is at rest is 10 N. The coefficient of static friction μ is

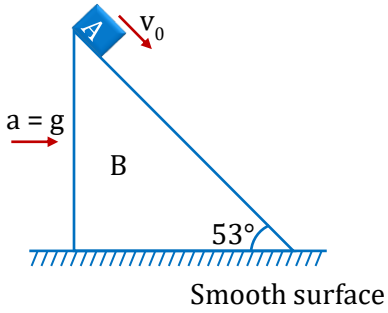


(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{\sqrt{6}}$ (C) $\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$

PNL150

For More Join: @IITJEE_Advanced

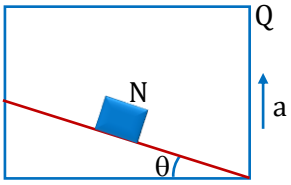
33. Block A is placed on a rough wedge B which is placed on a smooth surface. The wedge has angle of inclination of 53° and is imparted a horizontal acceleration g towards right. Block A is given an initial velocity v_0 with respect to wedge. Find the coefficient of friction for which block A moves with constant velocity v_0 with respect to wedge. ($g = 10\ m/s^2$)



(A) 2/3 (B) 3/7 (C) 1/7 (D) 1/3

PNL149

34. A block of mass m is at rest with respect to a rough incline kept in elevator moving up with acceleration a. Which of following statements is /are correct.

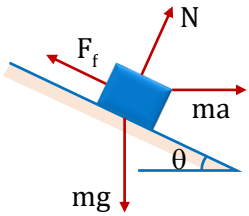


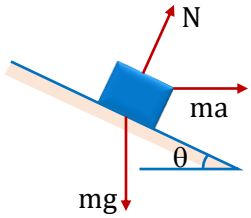
Bansers Academy JEE

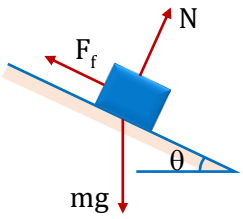
(A) The contact force between block and incline is vertical.
(B) The contact force between block and incline is of magnitude $m(g - a)$.
(C) The contact force between block and incline is perpendicular to the incline.
(D) The contact force is of magnitude $mg \cos\theta$.

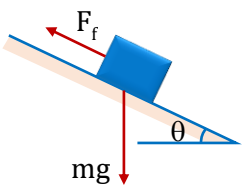
PNL152

35. A block rests on an inclined plane at an angle θ with horizontal and there is friction between the block and the plane. The plane is accelerated to the right. If the block remains at the same position on the plane, which of the following pictures might show the free-body diagram for the block? All observation is drawn for inertial observer (All of the vectors shown are forces).

(A)


(B)


(C)


(D)


PNL151

For More Join: @IITJEE_Advanced

36. Base angles of triangular wedge *ABC* are 37° and 90°. A uniform rope is kept over the wedge as shown in two figures. The maximum length of string which can be hanged vertically is x_1 (fig. I) and minimum length is x_2 (fig. II) in equilibrium. If $\frac{x_1}{x_2}=3$, find the value of coefficient of friction between rope and wedge.

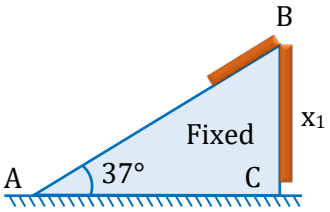


Fig. (i)

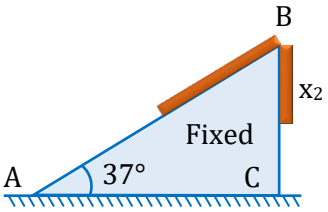
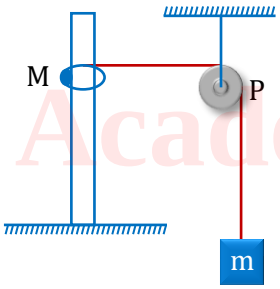


Fig. (ii)

(A) 0.5 (B) 0.4 (C) 0.6 (D) 0.8

PNL154

37. In the figure shown a ring of mass *M* and a block of mass *m* are in equilibrium. The string is light and pulley *P* does not offer any friction and coefficient of friction between pole and *M* is μ. The frictional force offered by the pole on *M* is



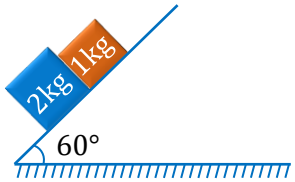
Rankers Academy JEE

(A) *Mg* directed up (B) μ*mg* directed up

(C) (*M*–*m*)*g* directed down (D) μ*mg* directed down

PNL013

38. In the figure shown if friction coefficient of block 1*kg* and 2*kg* with inclined plane is $\mu_1 = 0.5$ and $\mu_2 = 0.4$ respectively, then



(A) both blocks will move together.

(B) both blocks will move separately.

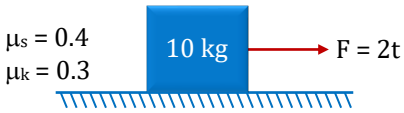
(C) there is a non-zero contact force between two blocks.

(D) none of these

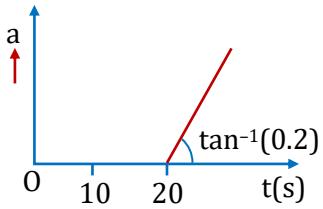
PNL019

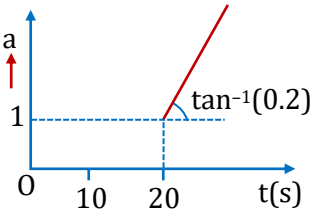
For More Join: @IITJEE_Advanced

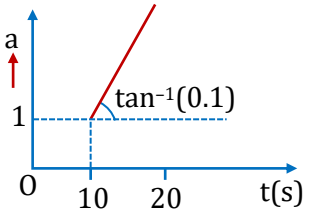
39. For shown situation the acceleration of block is represented by : (Here t is time in second) ($g = 10\ ms^{-2}$)



So, acceleration of block remains same.

(A)


(B)


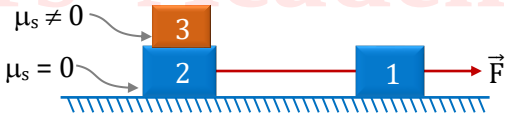
(C)


(D) None of these

PNL156

40. Three blocks of equal mass M are initially at rest on smooth floor. A force is applied to the system so that the three blocks are to move together. Mark the correct option.

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(A) The minimum coefficient of friction required is $\frac{F}{3Mg}$.

(B) The minimum coefficient of friction required is $\frac{F}{Mg}$.

(C) The minimum coefficient of friction required is $\frac{F}{2Mg}$.

(D) The minimum coefficient of friction required is $\frac{3F}{2Mg}$.

PNL158

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

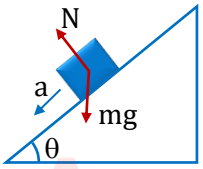
1. Consider a block suspended from a light string as shown in the figure.
Which of the following pairs of forces constitute Newton's third law pair?



(A) Force with which string pulls on the ceiling and the force with which string pulls on block
(B) Force with which string pulls on the block and weight of the block
(C) Force acting on block due to the earth and force the block exerts on the earth
(D) Force with which block pulls on string and force with which the string pulls on the block

PNL081

2. For the system shown which of the option(s) are **CORRECT** :



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(A) $N\cos\theta = mg$ (B) $N = mg\cos\theta$
(C) $N - mg\cos\theta = ma$ (D) $mg - N\cos\theta = ma\sin\theta$

PNL083

3. A carpenter of mass 50 kg is standing on a weighing machine placed in a lift of mass 20 kg. A light string is attached to the lift. The string passes over a smooth pulley and the other end is held by the carpenter as shown. When carpenter keeps the lift moving upward with constant velocity :-
($g = 10\ m/s^2$)

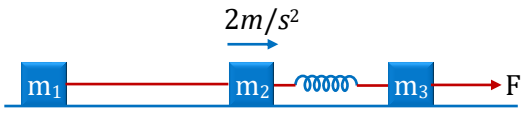


(A) the reading of weighing machine is 15 kg
(B) the man applies a force of 350 N on the string
(C) net force on the man is 150 N
(D) net force on the weighing machine is 150 N

PNL086

For More Join: @IITJEE_Advanced

4. In the given figure $m_1 = 1kg$, $m_2 = 2kg$, $m_3 = 3kg$, $F = 18\ N$ and spring constant is $100\ N/m$ then choose the correct options. (Assume string and spring are ideal)



(A) Acceleration of m_1 is $2\ m/s^2$

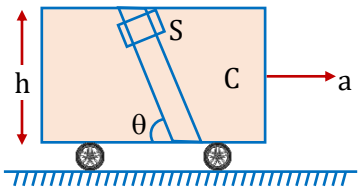
(B) Tension in string is $2N$

(C) Force applied by spring on block of mass m_2 is $6N$

(D) Acceleration of block of mass m_3 is $4\ m/s^2$

PNL097

5. A sleeve S of mass m starts sliding down along a smooth inclined rod through a vertical distance h from rest. Cart(C) carrying the rod is moving with constant horizontal acceleration a. If N is the normal force between sleeve and rod, $V_{S/C}$ is the speed of sleeve relative to cart on reaching the lowest point and T is the time taken to reach lowest point then :



(A) $N = m\ (g\cos\theta + a\sin\theta)$

(B) $V_{S/C} = \sqrt{2h(g - a\cot\theta)}$

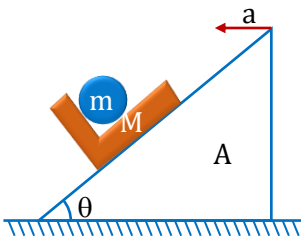
(C) $V_{S/C} = \sqrt{2h(g - a\tan\theta)}$

(D) $T = \sqrt{\dfrac{2h}{\sin\theta(g\cos\theta - a\sin\theta)}}$

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PNL112

6. A ball of mass m is placed on a sledge of mass M. The sledge moves on an inclined plane A of angle of inclination θ. Inclined plane A is moving with constant horizontal acceleration 'a' towards left (Neglect friction everywhere)



(A) Normal reaction force between m and M is $m(g\cos\theta + a\sin\theta)$

(B) Normal reaction force between A and M is $(M + m)\ (g\cos\theta + a\sin\theta)$

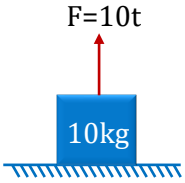
(C) $a_{M/A} = a_{m/A} = (g\sin\theta - a\cos\theta)$

(D) $a_M = a_m = \sin\theta\sqrt{g^2 + a^2}$

PNL120

For More Join: @IITJEE_Advanced

7. A block of $m = 10\ kg$ is kept on ground. A vertically upward force $F = (10\ t)N$, where t is the time in seconds starts acting on it at $t = 0$ as shown in figure.



(A) Time at which block loose contact from the surface is 10 sec.

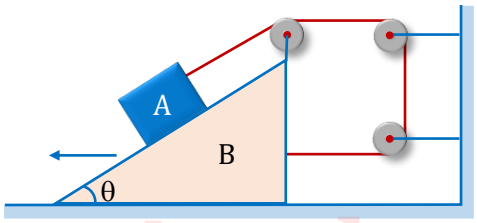
(B) Velocity of block at $t = 20$ sec is 50 m/s.

(C) Velocity of block at $t = 20$ sec is 200 m/s.

(D) Displacement of block in 20 sec is $\frac{500}{3}\ m$.

PNL192

8. A block A and wedge B connected through a string as shown. The wedge B is moving away from the wall with acceleration 2 m/s^2 horizontally and acceleration of block A is vertically upwards. Then



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(A) Acceleration of A with respect to B is 4 m/s^2.

(B) Acceleration of A with respect to B is $2\sqrt{3}\ m/s^2$.

(C) Angle θ is 60°.

(D) Acceleration of A is $2\sqrt{3}\ m/s^2$.

PNL134

9. A block is kept on a rough surface and applied with a horizontal force as shown which is gradually increasing from zero. The coefficient of static and kinetic friction are $\frac{1}{\sqrt{3}}$ then



(A) When F is less than the limiting friction, angle made by net force on the block by the surface is less than 30° with vertical.

(B) When the block is just about to move, the angle made by net force by the surface on the block becomes equal to 30° with vertical.

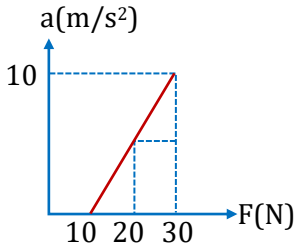
(C) When the block starts to accelerate, the angle made by net force by the surface on the block becomes constant and equal to 30° vertical.

(D) The angle made by net force with vertical on the block by the surface, depends on the mass of the block.

PNL062

For More Join: @IITJEE_Advanced

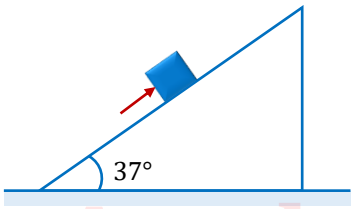
10. A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown.



(A) Mass of the block is 2 kg.
(B) Coefficient of friction between block and surface is 0.5.
(C) Limiting friction between block and surface is 10 N.
(D) When F = 8 N, friction between block and surface is 10 N.

PNL063

11. A block placed on rough inclined plane is pushed with a force F. The coefficient of friction is $\mu = 0.5$ and mass of block is 10 kg.



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(A) If F = 30 N, acceleration of block is zero and friction acting on block is 30 N.
(B) If F = 60 N, acceleration of block is zero and friction acting on block is 60 N.
(C) If F = 90 N, acceleration of block is zero and friction acting on block is 30 N.
(D) If F = 120 N, acceleration of block is 2 m/s^2 and friction acting on block is 40 N.

PNL194

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 12 TO 15

A uniform rope of mass (m) and length (L) placed on frictionless horizontal ground is being pulled by two forces F_A and F_B at its ends as shown in the figure. As a result, the rope accelerates toward the right.



12. Acceleration (a) of the rope is

(A) zero (B) $a = \frac{F_A+F_B}{m}$ (C) $a = \frac{F_A-F_B}{m}$ (D) $a = \frac{F_B-F_A}{m}$

PNL087

13. Tension (T) at the mid point of the rope is

(A) $T = F_B - F_A$ (B) $T = F_A + F_B$ (C) $T = \frac{1}{2}(F_B - F_A)$ (D) $T = \frac{1}{2}(F_A + F_B)$

PNL088

For More Join: @IITJEE_Advanced

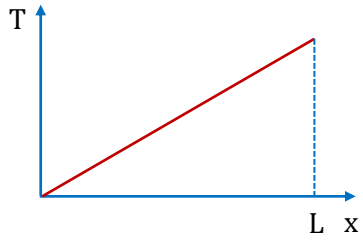
14. Expression (T_x) of tension at a point at distance x from the end A is

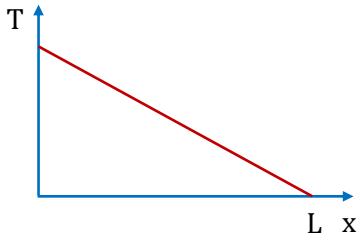
(A) $T_x = \left(\frac{F_B - F_A}{L}\right)x + F_A$ (B) $T_x = \left(\frac{F_B - F_A}{L}\right)x - F_A$

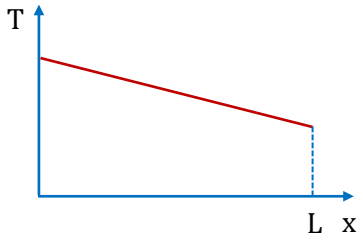
(C) $T_x = \left(\frac{F_B - F_A}{L}\right)x + F_B$ (D) $T_x = \left(\frac{F_B - F_A}{L}\right)x - F_B$

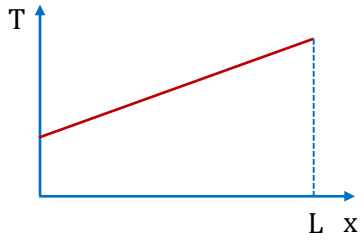
PNL089

15. Which of the following graph best represents variation in tension at a point on the rope with distance x of the point from the end A?

(A) 

(B) 

(C) 

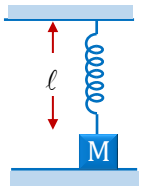
(D) 

PNL090

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PARAGRAPH FOR QUESTION NO. 16 TO 18

A block of mass m is placed on a smooth horizontal floor is attached to one end of spring. The other end of the spring is attached to fixed support. When spring is vertical it is relaxed. Now the block is pulled towards right by a force F, which is being increased gradually. When the spring makes angle 53° with the vertical, block leaves the floor.



16. When blocks leaves the table, the normal force on it from table is

(A) mg (B) zero (C) $\frac{4mg}{3}$ (D) $\frac{3mg}{4}$

PNL093

17. Force constant of the spring is :-

(A) $\frac{5mg}{2\ell}$ (B) $\frac{15mg}{8\ell}$ (C) $\frac{5mg}{3\ell}$ (D) $\frac{5mg}{4\ell}$

PNL094

18. When the block leaves the table, the force F is :-

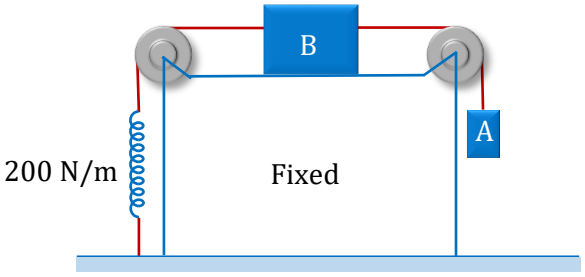
(A) $\frac{3mg}{4}$ (B) $\frac{4mg}{3}$ (C) $\frac{3mg}{5}$ (D) $\frac{4mg}{5}$

PNL095

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 19 TO 21

In the figure shown blocks A and B are of mass 2 kg and 8 kg and they are connected through strings to a spring connected to ground. The blocks are in equilibrium. ($g = 10 m/s^2$)



19. The elongation of the spring is

(A) 1 cm (B) 10 cm (C) 0.1 cm (D) 1m

PNL107

20. Now the block A is pulled downwards by a force gradually increasing to 20 N. The new elongation of spring is :-

(A) 2 cm (B) 4 cm (C) 20 cm (D) 40 cm

PNL108

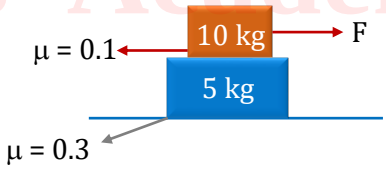
21. Now the force on A is suddenly removed. The acceleration of block B becomes :-

(A) 1.0 m/s (B) 2.0 m/s^2 (C) 3.0 m/s^2 (D) 4.0 m/s^2

PNL109

Rankers Academy JEE

PARAGRAPH FOR QUESTION NO. 22 TO 26



22. When $F = 2N$, the frictional force between 5 kg block and ground is

(A) 2 N (B) 0 (C) 8 N (D) 10 N

PNL160

23. When $F = 2N$, the frictional force between 10 kg block and 5 kg block is

(A) 2N (B) 15N (C) 10 N (D) None

PNL161

24. The maximum F which will not cause motion of any of the blocks is

(A) 10 N (B) 15 N (C) data insufficient (D) None

PNL162

25. The maximum acceleration of 5 kg block is :-

(A) 1 m/s^2 (B) 3 m/s^2 (C) 0 (D) None

PNL163

26. The acceleration of 10 kg block when $F = 30$ N is

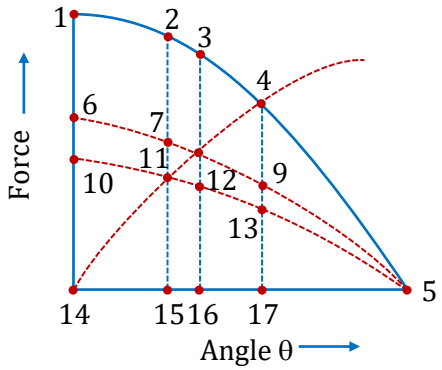
(A) 2 m/s^2 (B) 3 m/s^2 (C) 1 m/s^2 (D) None

PNL164

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 27 AND 28

Given graph is a plot of magnitude of normal, limiting friction, kinetic friction and static friction versus angle of inclination θ for the coin lying on book. 17 points are labeled with numbers. The coin slips at a certain angle θ_c. The coefficient of static friction is 0.6, and the coefficient of kinetic friction is 0.4.



27. The magnitude of the normal force on the coin from the book,

(A) 6 (B) 2 (C) 7 (D) 11

PNL170

28. θ_C is given by point

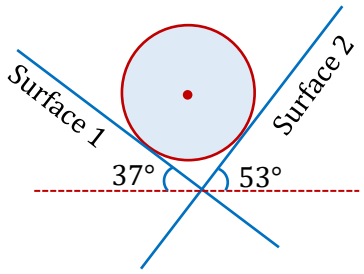
(A) 14 (B) 15 (C) 16 (D) 17

Rankers Academy JEE

PNL171

MATRIX MATCH TYPE QUESTION

29. A sphere of mass 10 *kg* is placed in equilibrium in a *V* shaped groove plane made of two smooth surfaces 1 and 2 as shown in figure. (g = 10 ms^{-2})

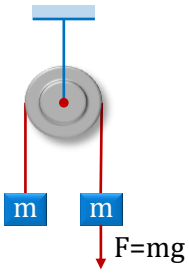
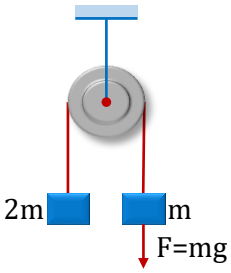
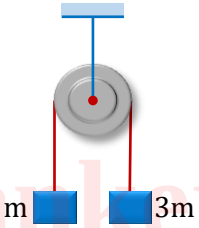
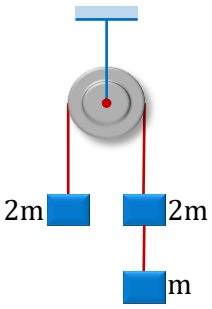


	Column-I		Column-II
(A)	Normal reaction by Surface 1	(P)	Zero
(B)	Normal reaction by surface 2	(Q)	60 *N*
(C)	Force on sphere by Earth	(R)	80 *N*
(D)	Net force on sphere	(S)	100 *N*
		(T)	120 *N*

PNL079

For More Join: @IITJEE_Advanced

30. Match the situations in column I to the accelerations of blocks in the column II (acceleration due to gravity is g and F is an additional force applied to one of the blocks ?

Column-I		**Column-II**	
(A)		(P)	$\frac{g}{5}$
(B)		(Q)	$\frac{g}{3}$
(C)		(R)	$\frac{g}{2}$
(D)		(S)	$\frac{2g}{3}$
		(T)	zero

PNL195

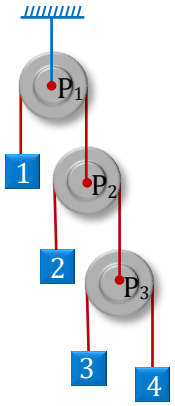
31. Velocity of three particles A, B and C varies with time t as, $\vec{v}_A = (2t\hat{\imath} + 6\hat{\jmath})\ m/s$; $\vec{v}_B = (3\hat{\imath} + 4\hat{\jmath})\ m/s$ and $\vec{v}_C = (6\hat{\imath} - 4t\hat{\jmath})\ m/s$. Regarding the pseudo force match the following table

	Column-I		**Column-II**
(A)	On A as observed by B	(P)	Along positive x-direction
(B)	On B as observed by C	(Q)	Along negative x-direction
(C)	On A as observed by C	(R)	Along positive y-direction
(D)	On C as observed by A	(S)	Along negative y-direction
		(T)	Zero

PNL119

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For More Join: @IITJEE_Advanced

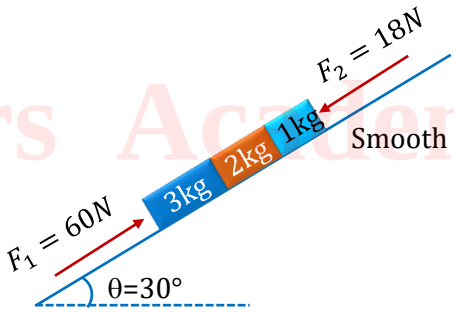
32. In the figure shown, acceleration of 1 is x (upwards). Acceleration of pulley P_3, w.r.t. pulley P_2 is y (downwards) and acceleration of 4 w.r.t. to pulley P_3 is z (upwards). Then



	Column-I		Column-II
(A)	Absolute acceleration of 2	(P)	$(y - x)$ downwards
(B)	Absolute acceleration of 3	(Q)	$(z - x - y)$ upwards
(C)	Absolute acceleration of 4	(R)	$(x + y + z)$ downwards
		(S)	None

PNL123

33. In the diagram shown in figure ($g = 10\ m/s^2$)



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	Column-I		Column-II
(A)	Acceleration of 2 kg block in m/s^2	(P)	8
(B)	Net force on 3 kg block in newton	(Q)	25
(C)	Normal reaction between 2 kg and 1 kg in newton	(R)	2
(D)	Normal reaction between 3 kg and 2 kg in newton	(S)	45
		(T)	None

PNL196

For More Join: @IITJEE_Advanced

EXERCISE (S)

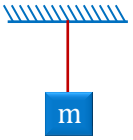
Newton's Laws of Motion

1. Draw free body diagram of block of mass *m*.

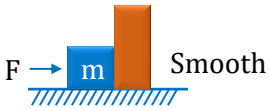
(a)



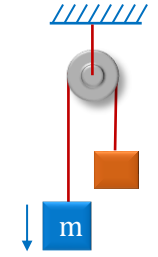
(b)



(c)



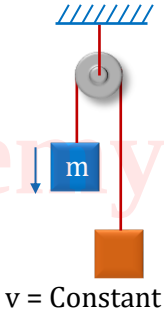
(d)



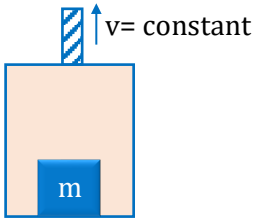
(e)



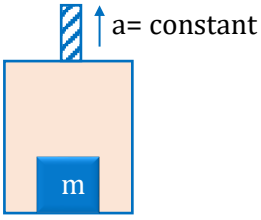
(f)



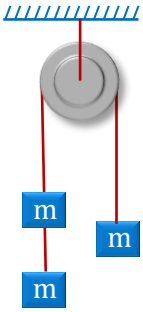
(g)



(h)



(i)

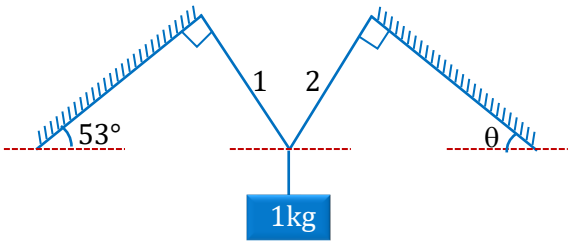


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PNL072

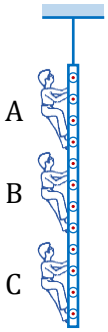
For More Join: @IITJEE_Advanced

2. In the given figure, what is the minimum possible tension (N) in the string 2 ?



PNL082

3. A ladder is hanging from ceiling as shown in figure. Three men A, B and C of masses 40 kg, 60 kg, and 50 kg are climbing the ladder. Man A is going up with retardation 2 m/s^2, C is going up with an acceleration of 1 m/s^2 and man B is going up with a constant speed of 0.5 m/s. Find the tension in the string supporting the ladder. [g = 9.8 m/s^2]

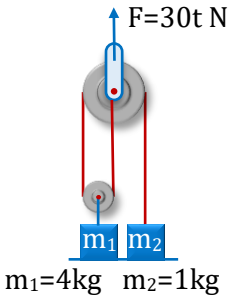


PNL197

4. A 40 kg monkey climbs a rope that passes over an ideal pulley. The other end of the rope is attached to a 60 kg weight placed on the ground. What is the maximum upward acceleration the monkey can have without lifting the weight? If he climbs the rope with upward acceleration 2g, with what acceleration the weight will rise up?

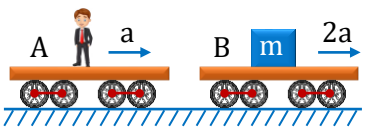
PNL085

5. Force F is applied on upper pulley. If F = 30t N where t is time in second. Find the time when m_1 loses contact with floor.



PNL198

6. Two trolley A and B are moving with accelerations a and 2a respectively in the same direction. To an observer in trolley A, the magnitude of pseudo force acting on a block of mass m on the trolley B is



PNL118

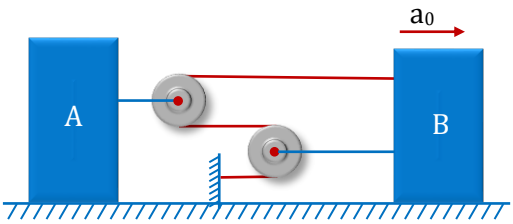
For More Join: @IITJEE_Advanced

7. A box of mass m is placed on a smooth horizontal platform as shown in the figure. The platform is made to move in direction 30° above the horizontal with acceleration a so that the contact force between the box and the platform becomes $3mg/2$. Find the magnitude of the acceleration (m/s^2).



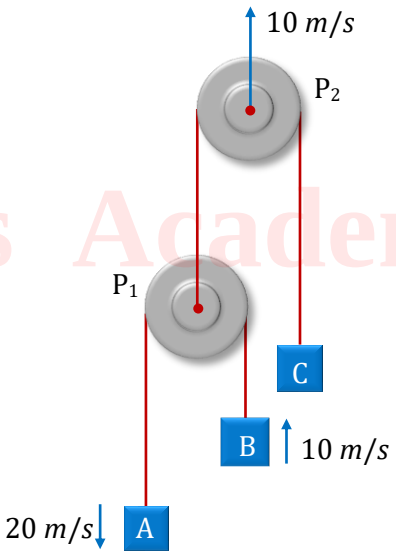
PNL069

8. Calculate the relative acceleration of A w.r.t. B if B is moving with acceleration a_0 towards right.



PNL122

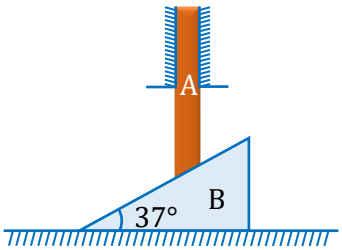
9. Velocities of blocks A, B and pulley p_2 are shown in figure. Find velocity of pulley p_1 and block C.



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PNL130

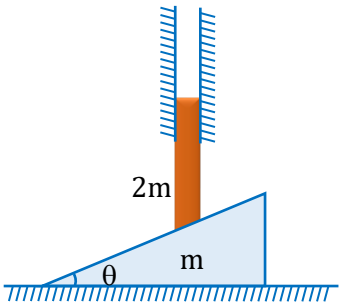
10. Find the acceleration of rod A and wedge B in the arrangement shown in figure if the mass of rod is equal to that of the wedge and the friction between all contact surfaces is negligible and rod A is free to move downwards only. Take angle of wedge as 37°.



PNL133

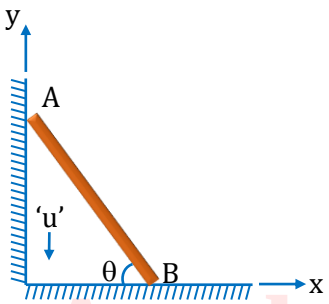
For More Join: @IITJEE_Advanced

11. A rod of mass $2m$ moves vertically downward on the surface of wedge of mass as shown in figure. Find the relation between velocity of rod and that of the wedge at any instant.



PNL132

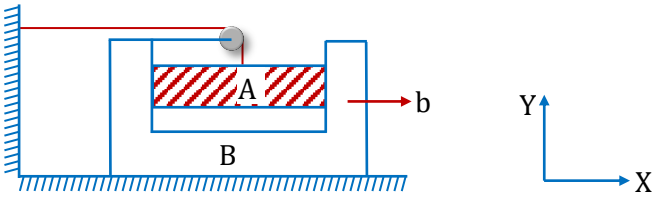
12. The velocity of end 'A' of rigid rod placed between two smooth vertical walls is 'u' along vertical direction. Find out the velocity of end 'B' of that rod, rod always remains in contact with the vertical wall.



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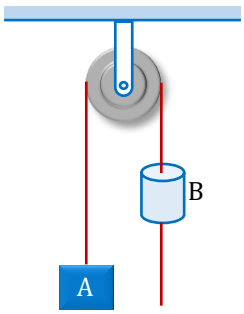
PNL136

13. Find the acceleration of A in term of b.



PNL137

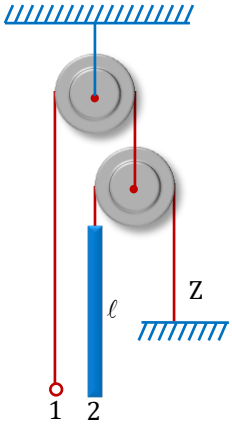
14. A 2 kg block A is attached to one end of a light string that passes over an ideal pulley and a 1 kg sleeve B slides down the other part of the string with an acceleration of 5 m/s^2 with respect to the string. Find the acceleration of the block, acceleration of sleeve and tension in the string.
[g = 10 m/s^2]



PNL200

For More Join: @IITJEE_Advanced

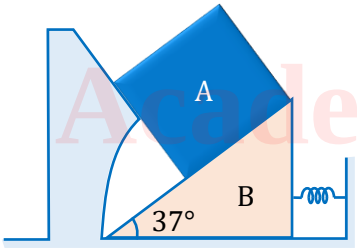
15. In the arrangement shown in figure the mass of ball is $\eta = 1.8$ times as great as that of rod 2. The length of the latter is $l = 100\ cm$. The masses of the pulleys and the threads, as well as the friction, are negligible. The ball is set on the same level as the lower end of the rod and then released. How soon will the ball be opposite the upper end of the rod?



PNL201

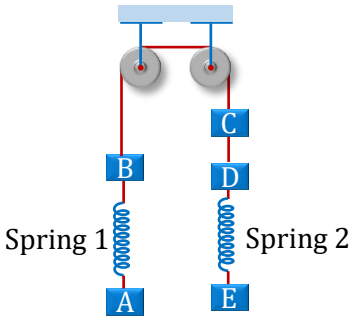
16. In the figure shown, all surfaces are smooth and block A and wedge B have mass 10 kg and 20 kg respectively. Find normal reaction between block A and B, spring force and normal reaction of ground on block B. ($g = 10\ m/s^2$).

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PNL092

17. The system shown in the figure is initially in equilibrium. A is of mass $2m$ and B, C, D and E are of mass m. Certain actions are performed on the system. Every action has been taken individually when the system is intact. Find the direction and magnitude of acceleration of the blocks after each of the following actions has been taken:



(i) Spring 1 is cut

(ii) Spring 2 is cut

(iii) String between C and D is cut.

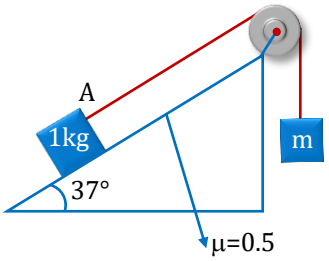
(iv) String between B and C is cut.

PNL106

For More Join: @IITJEE_Advanced

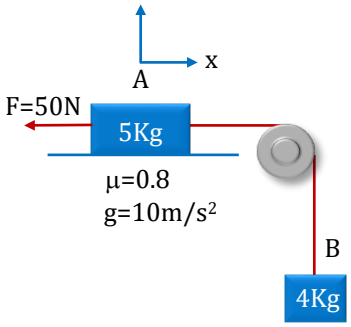
Friction

18. In the figure, what should be mass *m* so that block *A* slides up with a constant velocity?



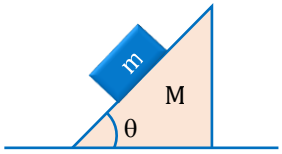
PNL202

19. Find the acceleration of the blocks and magnitude and direction of frictional force between block *A* and table, if block *A* is pulled towards left with a force of 50 *N*.



PNL203

20. A block of mass *m* lies on wedge of mass *M* as shown in figure. Find the minimum friction coefficient required between wedge *M* and ground so that it does not move while block *m* slips down on it.

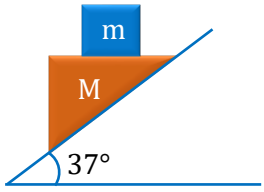


PNL172

21. You pour a quantity of flour of volume $V = 72\ cm^3$ onto a board, where it forms a conical pile. The coefficient of static friction between flour grains is $\mu_s = \sqrt{\pi}$. Find the maximum height (in *cm*) of the pile.

PNL169

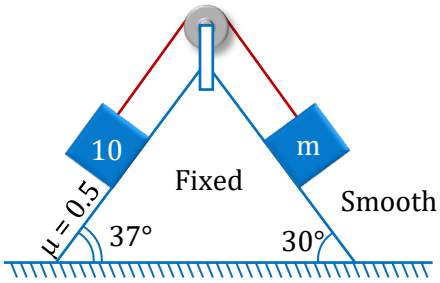
22. Block *M* slides down on frictionless incline as shown. The minimum friction coefficient so that *m* does not slide with respect to *M* is $\frac{n}{4}$, find the value of *n*.



PNL030

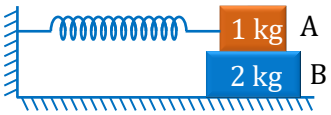
Rankers Academy JEE
For More Join: @IITJEE_Advanced

23. Find minimum value of m so that 10 kg block moves up the inclined plane.



PNL155

24. In the arrangement shown coefficient of friction between the two blocks A and B is 0.6 and ground is frictionless. Force constant of spring is 900 N/m. Find the maximum elongation of spring (in cm) such that the blocks do not slip over each other when released.



PNL168

25. Coefficient of friction between 5 kg and 10 kg block is 0.5. If friction between them is 20 N. What is the value of force being applied on 5 kg. The floor is frictionless.



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PNL165

For More Join: @IITJEE_Advanced

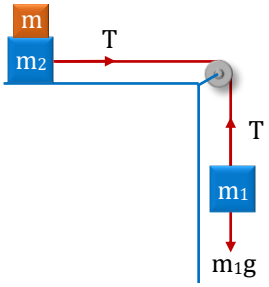
EXERCISE - JEE (Main) PYQ

1. A rocket is fired vertically from the earth with an acceleration of $2g$, where g is the gravitational acceleration. On an inclined plane inside the rocket, making an angle θ with the horizontal, a point object of mass m is kept. The minimum coefficient of friction μ_{min} between the mass and the inclined surface such that the mass does not move is: **[JEE (Main) 2016]**

(1) $2\tan\theta$ (2) $3\tan\theta$ (3) $\tan\theta$ (4) $\tan 2\theta$

PNL033

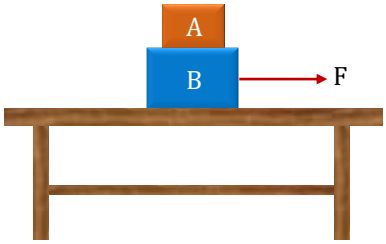
2. Two masses $m_1 = 5\ kg$ and $m_2 = 10\ kg$, connected by an inextensible string over a frictionless pulley, are moving as shown in the figure. The coefficient of friction of horizontal surface is 0.15. The minimum weight m that should be put on top of m_2 to stop the motion is :- **[JEE (Main) 2018]**



(1) $27.3\ kg$ (2) $43.3\ kg$ (3) $10.3\ kg$ (4) $18.3\ kg$

PNL034

3. Two blocks A and B of masses $m_A = 1\ kg$ and $m_B = 3$ kg are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2. The maximum force F that can be applied on B horizontally, so that the block A does not slide over the block B is : (Take $g = 10\ m/s^2$) **[JEE (Main) 2019]**



(1) $16N$ (2) $40N$ (3) $12N$ (4) $8N$

PNL035

4. A ball is thrown upward with an initial velocity V_0 from the surface of the earth. The motion of the ball is affected by a drag force equal to $m\gamma\upsilon^2$ (where m is mass of the ball, υ is its instantaneous velocity and γ is a constant). Time taken by the ball to rise to its zenith is **[JEE (Main) 2019]**

(1) $\frac{1}{\sqrt{\gamma g}}\sin^{-1}\left(\sqrt{\frac{\gamma}{g}}V_0\right)$ (2) $\frac{1}{\sqrt{\gamma g}}\tan^{-1}\left(\sqrt{\frac{\gamma}{g}}V_0\right)$

(3) $\frac{1}{\sqrt{2\gamma g}}\tan^{-1}\left(\sqrt{\frac{2\gamma}{g}}V_0\right)$ (4) $\frac{1}{\sqrt{\gamma g}}\ln\left(1+\sqrt{\frac{\gamma}{g}}V_0\right)$

PNL036

For More Join: @IITJEE_Advanced

5. A spring whose unstretched length is l has a force constant k. The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constants, k_1 and k_2 will be : **[JEE (Main) 2019]**

(1) $\frac{1}{n^2}$ (2) n^2 (3) $\frac{1}{n}$ (4) n

PNL037

6. An insect is at the bottom of a hemispherical ditch of radius $1m$. It crawls up the ditch but starts slipping after it is at height h from the bottom. If the coefficient of friction between the ground and the insect is 0.75, then h is : ($g = 10\ ms^{-2}$) **[JEE (Main) 2020]**

(1) 0.80 m (2) 0.60 m (3) 0.45 m (4) 0.20 m

PNL039

7. A particle of mass M originally at rest is subjected to a force whose direction is constant but magnitude varies with time according to the relation

$$F = F_0\left[1-\left(\frac{t-T}{T}\right)^2\right]$$

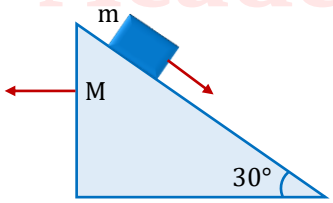
Where F_0 and T are constants. The force acts only for the time interval $2T$. The velocity v of the particle after time $2T$ is : **[JEE (Main) 2021]**

(1) $2F_0T/M$ (2) $F_0T/2M$ (3) $4F_0T/3M$ (4) $F_0T/3M$

PNL041

8. A block of mass m slides on the wooden wedge, which in turn slides backward on the horizontal surface. The acceleration of the block with respect to the wedge is : Given $m = 8\ kg$, $M = 16\ kg$ Assume all the surfaces shown in the figure to be frictionless. **[JEE (Main) 2021]**

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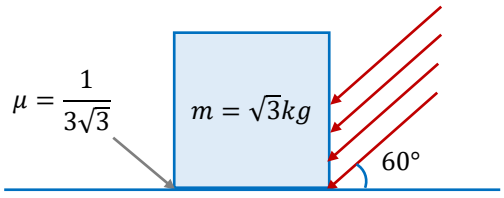


(1) $\frac{4}{3}g$ (2) $\frac{6}{5}g$ (3) $\frac{3}{5}g$ (4) $\frac{2}{3}g$

PNL042

9. As shown in the figure, a block of mass $\sqrt{3}\ kg$ is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move, will be $3x$. The value of x will be

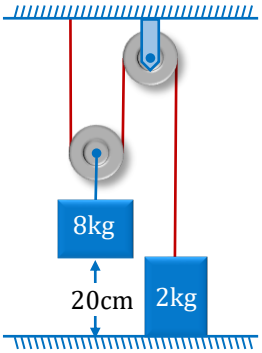
[$g = 10\ m/s^2$; $\sin 60° = \frac{\sqrt{3}}{2}$; $\cos 60° = \frac{1}{2}$] **[JEE (Main) 2021]**



PNL044

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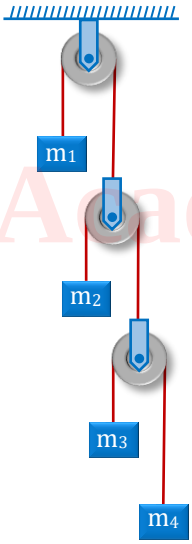
10. The boxes of masses 2 *kg* and 8 *kg* are connected by a massless string passing over smooth pulleys. Calculate the time taken by box of mass 8 *kg* to strike the ground starting from rest. (Use $g = 10\ m/s^2$) **[JEE (Main) 2021]**



(1) 0.34 *s* (2) 0.2 *s* (3) 0.25 *s* (4) 0.4 *s*

PNL048

11. In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement? **[JEE (Main) 2022]**

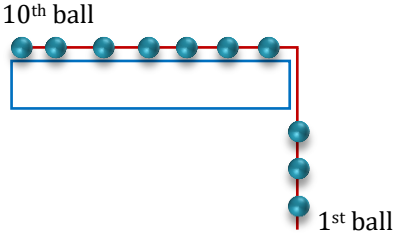


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(1) $4a_1 + 2a_2 + a_3 + a_4 = 0$ (2) $a_1 + 4a_2 + 3a_3 + a_4 = 0$
(3) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$ (4) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

PNL049

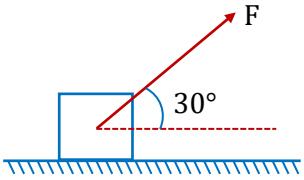
12. A system to 10 balls each of mass 2 *kg* are connected via massless and unstretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the 7th and 8th ball is_____*N* when 6th ball just leaves the table. **[JEE (Main) 2022]**



PNL050

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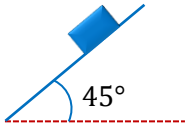
13. As shown in the figure a block of mass 10 *kg* lying on a horizontal surface is pulled by a force *F* acting at an angle 30°, with horizontal. For $\mu_s = 0.25$, the block will just start to move for the value of *F*: [Given g = 10 ms^{-2}] **[JEE (Main) 2023]**



(1) 33.3 *N* (2) 25.2 *N* (3) 20 *N* (4) 35.7 *N*

PNL210

14. Consider a block kept on an inclined plane (inclined at 45°) as shown in the figure. If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down, the coefficient of friction between the block and inclined plane (μ) is equal to : **[JEE (Main) 2023]**



(1) 0.33 (2) 0.60 (3) 0.25 (4) 0.50

PNL211

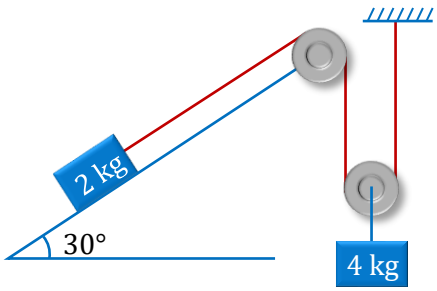
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15. A wooden block of mass 5 *kg* rests on soft horizontal floor. When an iron cylinder of mass 25 *kg* is placed on the top of the block, the floor yields and the block and the cylinder together go down with an acceleration of 0.1 ms^{-2}. The action force of the system on the floor is equal to: **[JEE (Main) 2024]**

(1) 297 *N* (2) 294 *N* (3) 291 *N* (4) 196 *N*

PNL261

16. All surfaces shown in figure are assumed to be frictionless and the pulleys and the string are light. The acceleration of the block of mass 2 *kg* is :



[JEE (Main) 2024]

(1) g (2) $\frac{g}{3}$ (3) $\frac{g}{2}$ (4) $\frac{g}{4}$

PNL262

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17. A block of mass m is placed on a surface having vertical cross section given by $y = x^2/4$. If coefficient of friction is 0.5, the maximum height above the ground at which block can be placed without slipping is: **[JEE (Main) 2024]**

(1) $1/4\ m$ (2) $1/2\ m$ (3) $1/6\ m$ (4) $1/3\ m$

PNL263

18. A given object takes n times the time to slide down 45° rough inclined plane as it takes the time to slide down an identical perfectly smooth 45° inclined plane. The coefficient of kinetic friction between the object and the surface of inclined plane is : **[JEE (Main) 2024]**

(1) $1-\frac{1}{n^2}$ (2) $1 - n^2$ (3) $\sqrt{1-\frac{1}{n^2}}$ (4) $\sqrt{1-n^2}$

PNL264

19. A massless spring gets elongated by amount x_1 under a tension of $5N$. Its elongation is x_2 under the tension of $7N$. For the elongation of $(5x_1 - 2x_2)$, the tension in the spring will be,

[JEE (Main) 2025]

(1) $15\ N$ (2) $20\ N$ (3) $11\ N$ (4) $39\ N$

PNL326

20. A balloon and its content having mass M is moving up with an acceleration 'a'. The mass that must be released from the content so that the balloon starts moving up with an acceleration '$3a$' will be: (Take 'g' as acceleration due to gravity) **[JEE (Main) 2025]**

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(1) $\frac{3Ma}{2a-g}$ (2) $\frac{3Ma}{2a+g}$ (3) $\frac{2Ma}{3a+g}$ (4) $\frac{2Ma}{3a-g}$

PNL327

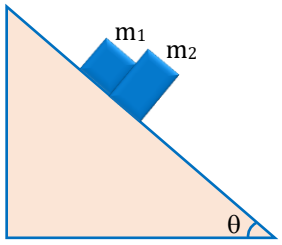
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EXERCISE - JEE (Advanced) PYQ

1. A block of mass $m_1 = 1kg$ another mass $m_2 = 2kg$, are placed together (see figure) on an inclined plane with angle of inclination θ. Various values of θ are given in List-I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to μ = 0.3. In List-II expressions for the friction on block m_2 are given. Match the correct expression of the friction in List-II with the angles given in List-I and choose the correct option. The acceleration due to gravity is denoted by g.

[**Useful information :** tan (5.5°) ≈ 0.1 ; tan(11.5°) ≈ 0.2 ; tan (16.5°) ≈ 0.3]

[JEE (Advanced) 2014]



List–I		**List–II**	
(P)	θ = 5°	(1)	$m_2g \sin\theta$
(Q)	θ = 10°	(2)	$(m_1 + m_2)g \sin\theta$
(R)	θ = 15°	(3)	$\mu m_2 g \cos\theta$
(S)	θ = 20°	(4)	$\mu(m_1 + m_2)g \cos\theta$

Code:

(A) P-1, Q-1, R-1, S-3 (B) P-2. Q-2, R-2, S-3

(C) P-2, Q-2, R-2, S-4 (D) P-2, Q-2, R-3, S-3

PNL071

2. A particle of mass m is moving in the xy-plane such that its velocity at a point (x, y) is given as $\vec{v} = \alpha(y\hat{x} + 2x\hat{y})$, where α is a non-zero constant. What is the force $\vec{F}$ acting on the particle ?

[JEE (Advanced) 2023]

(A) $\vec{F} = 2m\alpha^2(x\hat{x} + y\hat{y})$ (B) $\vec{F} = m\alpha^2(y\hat{x} + 2x\hat{y})$

(C) $\vec{F} = 2m\alpha^2(y\hat{x} + x\hat{y})$ (D) $\vec{F} = m\alpha^2(x\hat{x} + 2y\hat{y})$

PNL212

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	D	B	D	A	C	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	B	D	C	B	B	B	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	C	C	B	C	D	A	B	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	A	C	A	C	A	A	B	B	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	CD	BD	AB	ABCD	AB	ABCD	ABD	ACD	ABCD	ABC
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	ACD	D	D	A	D	B	A	B	B	C
Que.	21	22	23	24	25	26	27	28	29	
Ans.	B	A	A	A	C	A	B	C	(A-R), (B-Q), (C-S), (D-P)	
Que.	30		31		32		33			
Ans.	(A-R), (B-T), (C-R), (D-P)		(A-T), (B-R), (C-R), (D-Q)		(A-S), (B-R), (C-Q)		(A-R), (B-T), (C-Q), (D-T)			

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EXERCISE (S)

1. -
2. 8
3. 1440 N
4. 0.5 g, g
5. 2 sec
6. ma
7. 10
8. $\left(\frac{a_0}{2}\right)$
9. V_{P_1} = 5 m/s downward, V_C = 25 m/s upward
10. $a_A = \frac{9g}{25}$, $a_B = \frac{12g}{25}$
11. $u = v\tan\theta$ (u is velocity of rod, v is the velocity of wedge)
12. $u\tan\theta$
13. $(b\hat{i} + b\hat{j})$
14. 5 m/s^2 downwards, 0 m/s^2, 10 N
15. $t = \sqrt{2\ell(4+\eta)/3g(2-\eta)} = 1.4s$
16. 80 N, 48 N, 264 N
17. (i) $a_A = g\downarrow, a_B = \frac{2g}{3}\uparrow, a_C = \frac{2g}{3}\downarrow, a_D = \frac{2g}{3}\downarrow, a_E = 0$
(ii) $a_A = 0, a_B = \frac{g}{3}\downarrow, a_C = \frac{g}{3}\uparrow, a_D = \frac{g}{3}\uparrow, a_E = g\downarrow,$
(iii) $a_A = 0, a_B = g\downarrow, a_C = g\uparrow, a_D = 2g\downarrow, a_E = 0$
(iv) $a_A = 0, a_B = 3g\downarrow, a_C = \frac{3g}{2}\downarrow, a_D = \frac{3g}{2}\downarrow, a_E = 0$
18. 1 kg
19. $10\hat{i}$
20. $\left(\mu_{min} = \frac{m\sin\theta\cos\theta}{m\cos^2\theta + M}\right)$
21. 6
22. 3
23. 20 kg
24. 1
25. 30 N

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	1	2	3	4	3	4	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	36	2	1	3	2	1	1	3	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2
Ans.	D	A

Rankers Academy JEE

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